

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
(CPR) | IMCQ 06

CPR IMCQ 6 Answer Key

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Anatomy

1- A 35-year-old man comes to the emergency department because of a 2-day history of progressively worsening fever, chills, malaise, and a persistent cough. He also has sharp, intense pain during inspiration. He is a non-smoker, and has occasional seasonal allergies. He is diaphoretic upon arrival, and his vital signs show tachycardia and tachypnea, with blood pressure within the normal range. The oxygen saturation is slightly below normal limits. Physical examination shows dullness to percussion and decreased breath sounds are noticed over the affected lung region. Laboratory studies show a high white blood cell count. An x-ray of the chest is attached. Which of the following nerves is most likely responsible for the afferent signals coming from the affected part of the pleura?

- A. Intercostal nerves of the left hemithorax
- B. Phrenic Nerve
- C. Left vagus
- D. Intercostal nerves of the right hemithorax
- E. Sympathetics from cardiopulmonary plexus
- F. Visceral afferents



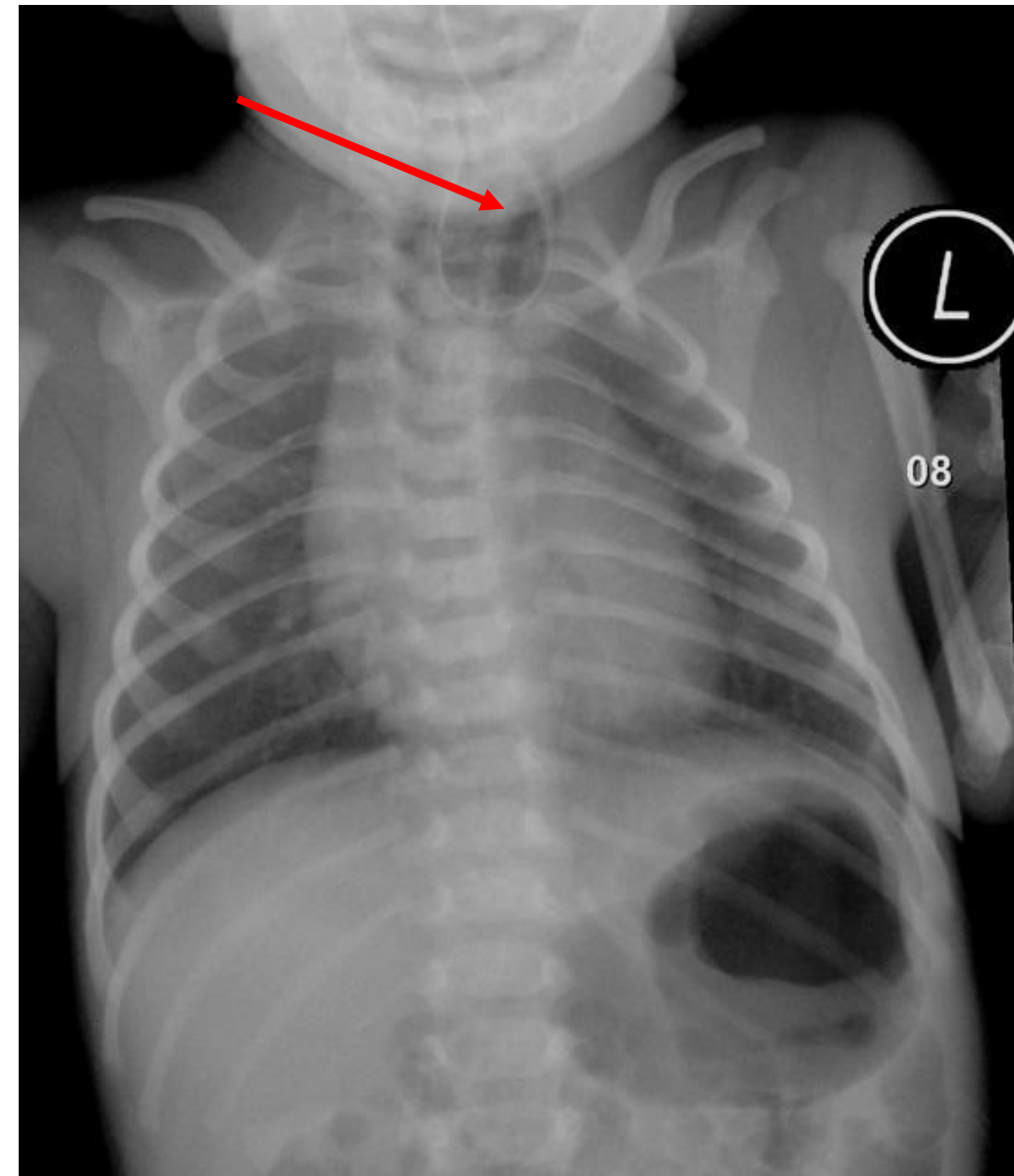
1- A 35-year-old man comes to the emergency department because of a 2-day history of progressively worsening fever, chills, malaise, and a persistent cough. He also has sharp, intense pain during inspiration. He is a non-smoker, and has occasional seasonal allergies. He is diaphoretic upon arrival, and his vital signs show tachycardia and tachypnea, with blood pressure within the normal range. The oxygen saturation is slightly below normal limits. Physical examination shows dullness to percussion and decreased breath sounds are noticed over the affected lung region. Laboratory studies show a high white blood cell count. An x-ray of the chest is attached. Which of the following nerves is most likely responsible for the afferent signals coming from the affected part of the pleura?

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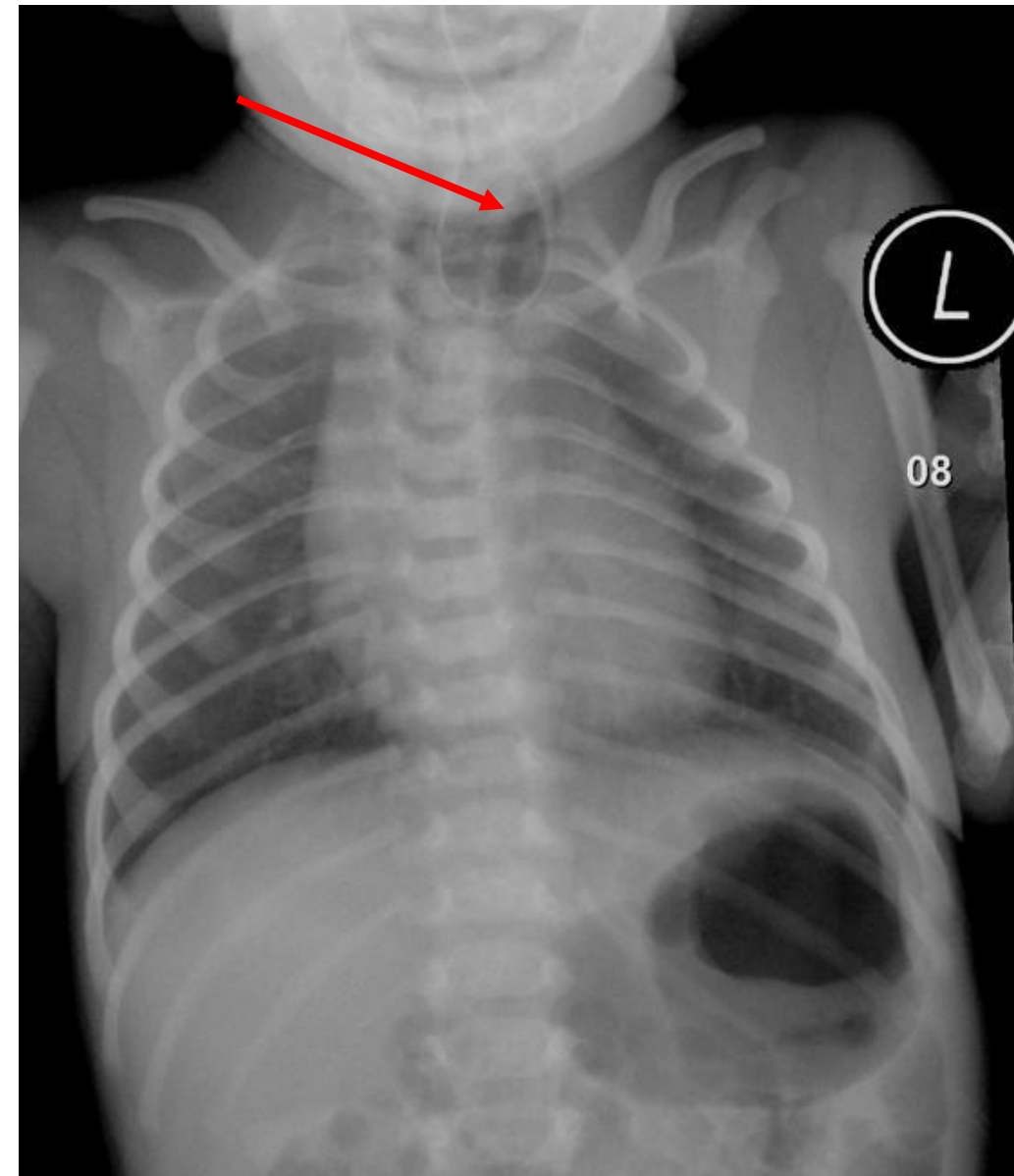
2- A newborn girl is evaluated after she chokes and coughs during her first feeding. Breast milk regurgitates from her mouth and nose. Noticing the difficulty of feeding, the physician tries to pass a catheter into her stomach, which is unsuccessful. A radiograph taken during catheterization shows the location of the radiopaque catheter (at the tip of the arrow). There is a bubble of gas seen in the stomach. Which of the following options best describes the embryological defect seen?

- A. Distal TEF with atretic proximal esophageal part
- B. Isolated esophageal stenosis with polyhydramnios
- C. Isolated esophageal atresia with polyhydramnios
- D. H-shaped Tracheoesophageal fistula with oligohydramnios
- E. Proximal TEF with atretic distal esophageal part
- F. Esophageal stenosis with oligohydramnios
- G. Respiratory distress syndrome with oligohydramnios



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3- A 3-year-old boy is brought to the emergency department by his parents because he suddenly began coughing and wheezing while playing with small plastic beads 20 minutes ago. He has no fever and was previously healthy. His temperature is 36.9°C (98.4°F), pulse is 118/min, respirations are 32/min, and blood pressure is 92/58 mm Hg. Physical examination shows increased work of breathing and decreased air entry over the right lower lung field. A chest radiograph shows hyperinflation of the right lung consistent with a ball-valve obstruction. Which of the following locations is the aspirated foreign body most likely lodged in?

- A. Left main bronchus
- B. Left secondary bronchus
- C. Right main bronchus
- D. Right secondary bronchus
- E. Trachea at the level of the cricoid cartilage

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4- A 2-day-old boy is brought to the emergency department by his mother because of a 1-day history of coughing, choking, and episodes of cyanosis after breastfeeding. He was born at term via an uncomplicated vaginal delivery, and the prenatal history shows no known maternal infections or complications during pregnancy. The mother reports that the baby had difficulty swallowing and showed signs of respiratory distress shortly after the first feeding. Physical examination shows the boy is tachypneic, has subcostal retractions, and a gurgling sound during respiration. A nasogastric tube is inserted but cannot pass beyond the upper esophagus. A radiograph taken during catheterization shows the location of the radiopaque catheter (at the tip of the arrow). What is the most likely diagnosis ?

- A. Esophageal atresia with distal fistula
- B. Proximal TEF with atretic distal esophageal part
- C. H shape TEF
- D. Tracheal atresia
- E. Atresia with double fistula
- F. Respiratory distress syndrome



4- A 2-day-old boy is brought to the emergency department by his mother because of a 1-day history of coughing, choking, and episodes of cyanosis after breastfeeding. He was born at term via an uncomplicated vaginal delivery, and the prenatal history shows no known maternal infections or complications during pregnancy. The mother reports that the baby had difficulty swallowing and showed signs of respiratory distress shortly after the first feeding. Physical examination shows the boy is tachypneic, has subcostal retractions, and a gurgling sound during respiration. A nasogastric tube is inserted but cannot pass beyond the upper esophagus. A radiograph taken during catheterization shows the location of the radiopaque catheter (at the tip of the arrow). What is the most likely diagnosis ?

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5- A 1-day-old neonate is examined by a physician because of a 2-hour history of extreme difficulty breathing. The boy was delivered at full-term and has been feeding well until this recent turn of events. He has been crying and appears uncomfortable, which the mother attributes to his breathing problems. Physical examination shows signs of respiratory distress and severe dyspnea. His breathing is rapid, labored, and accompanied by intercostal retractions. However, his pulse is elevated, and his oxygen saturation levels are slightly low. A chest x-ray is shown. Failure of which of the following processes is most likely the cause of his condition?

- A. Septum transversum fusion with the dorsal mesentery.
- B. Pleuroperitoneal folds fusion
- C. Septum transversum fusion with the costal margin on the right side
- D. Migration of lateral body wall muscles
- E. Pleuropericardial membranes fusion with the dorsal mesentery and the pleuroperitoneal membranes



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6. A 42-year-old woman comes to the emergency room because of a 4-day history of chest tightness and a dull ache behind her sternum. Her temperature is 37.1°C (98.8°F), pulse is 92/min, respirations are 18/min, and blood pressure is 128/76 mm Hg. Physical examination shows sharp pain with deep breaths but no respiratory distress. A CT scan of the chest shows an 8-cm mediastinal mass. A chest x-ray of the chest is shown. Which of the following nerves is most likely involved in the transmission of pain experienced in this patient?

- A. Accessory
- B. Intercostal
- C. Long thoracic
- D. Phrenic
- E. Vagus



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Histology Cell Biology Questions

7. A 54-year-old man comes to the physician because of a 3-month history of pain and swelling to his left foot from an embedded wooden splinter. His individual attempts to remove the splinter were unsuccessful. Physical examination shows an erythematous lump on the medial plantar aspect of his left foot. Microscopic examination of the excised lump shows the presence of multi-nucleated giant cells with centrally grouped nuclei. Which of the following best describes the precursor cells responsible for the formation of the giant cell?

- A. They are also called polymorphonuclear cells
- B. They contain granules with major basic protein
- C. They originate from the lymphoid lineage
- D. Are found in increased numbers 24-48hrs after onset of inflammation
- E. Have euchromatic nuclei and large Golgi apparatus

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8. A 75-year-old man is brought to the emergency department because of a 2-day history of change in mental status. His blood pressure is 100/70 mmHg, pulse is 104/min and temperature is 104°F (40°C). Physical examination shows a febrile, disoriented man with a fluctuant, tender mass in the right perianal area. The abscess is incised with drainage of an abundance of purulent material. Laboratories studies show markedly elevated neutrophils in blood. The physician suspects septicemia secondary to a perianal abscess. Which of the following mediators is most likely responsible for the abnormal temperature observed in this case?

- A. Interleukin 1 (IL-1)
- B. Histamine
- C. Nitric oxide
- D. C3a
- E. Serotonin

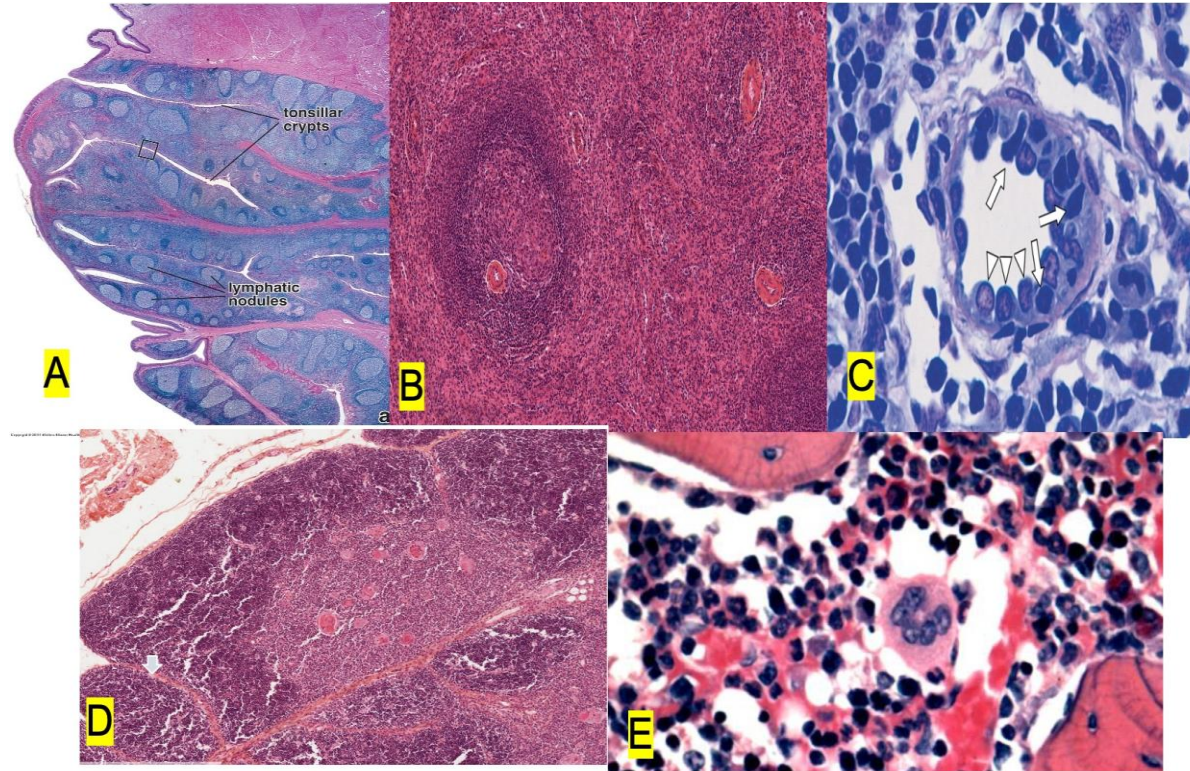
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SOM.MK.I.BPM1.3.CPR.1.HCB.1292 List the chemical mediators involved in various stages of acute inflammatory response.

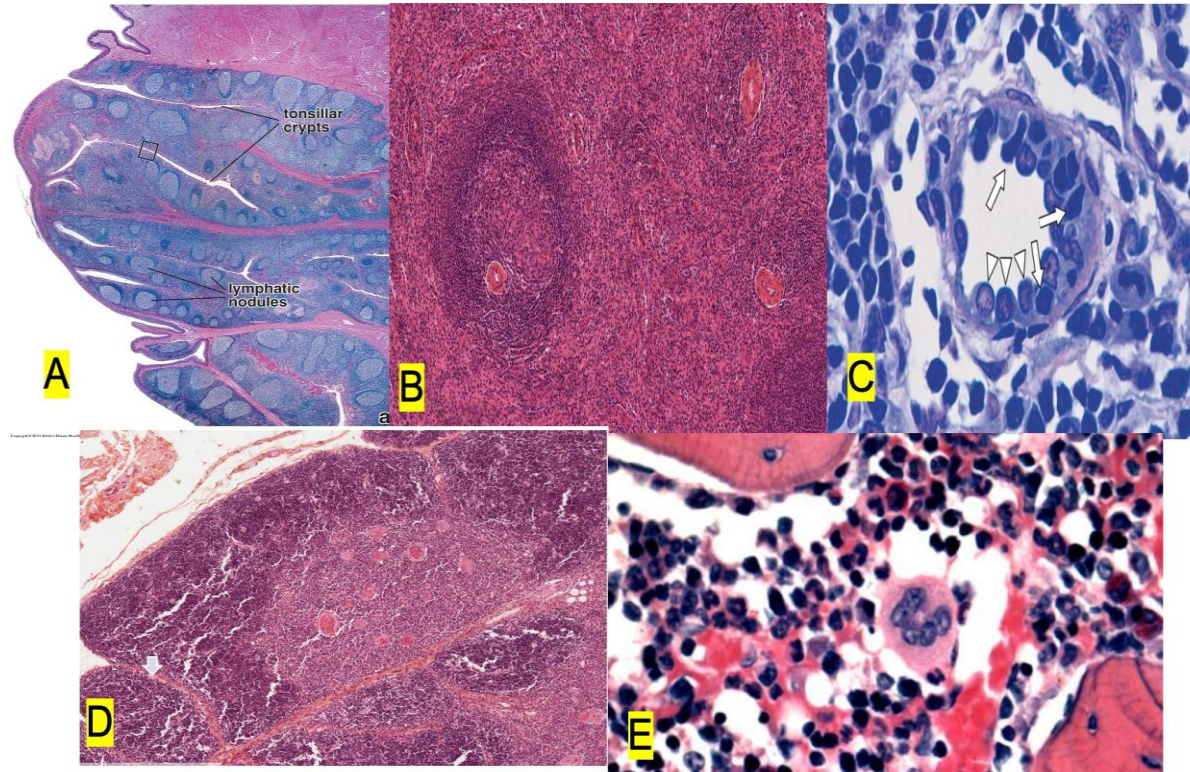
9. A 5-year-old girl is brought to the emergency department by her mother because of a 2-day history of fever and decreased appetite. Physical examination shows enlarged, erythematous palatine tonsils which are partially obscuring the oropharynx and are covered by patches of yellow exudate. Which of the following photomicrographs best represents the affected tissue?

- A. A
- B. B
- C. C
- D. D
- E. E



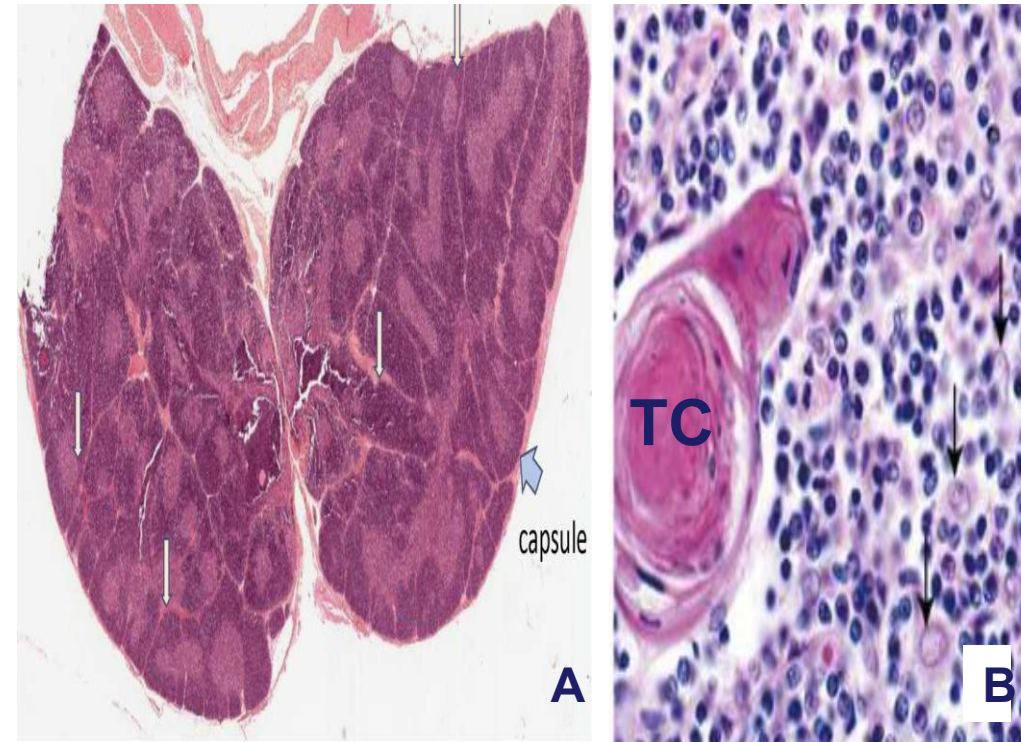
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- B. ☐ B
- C. ☐ C
- D. ☐ D
- E. ☐ E



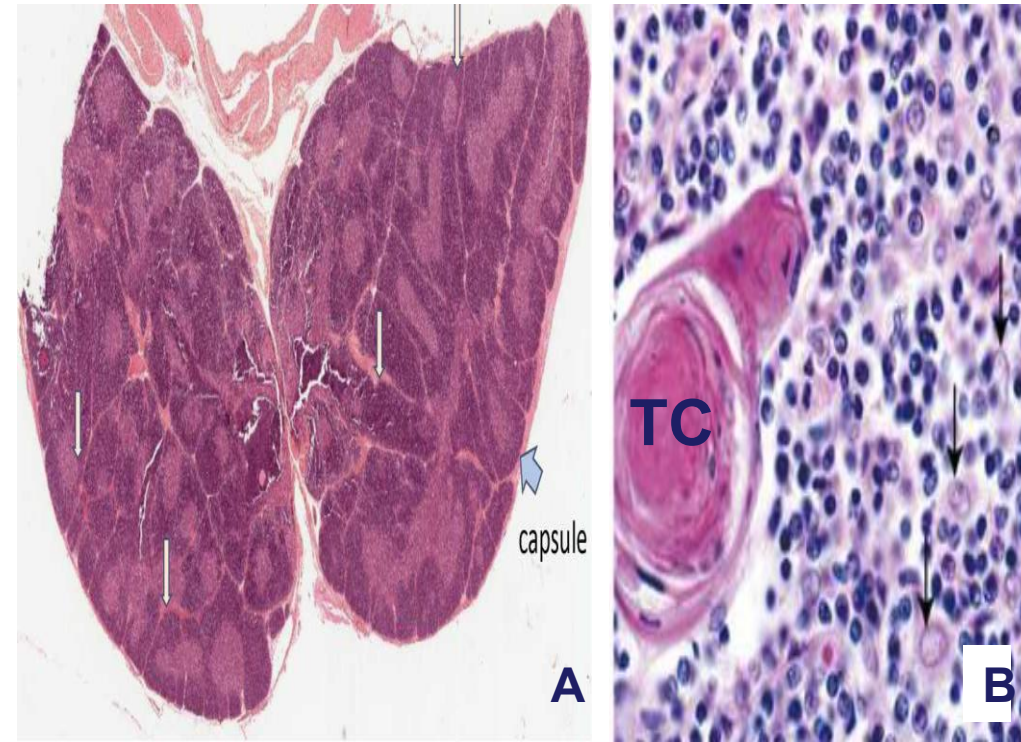
SOM.MK.1.BPM1.3.CPR.1.HCB.1245 Distinguish between the three (3) types of tonsils based on their location and microscopic structure.

10. A 14-year-old girl is brought to the physician by her parents because of a 2-month history of pain and difficulty swallowing. Physical examination shows visible veins in the neck. Imaging studies show enlargement of the organ shown in photomicrograph A. Microscopic examination of a biopsy specimen shows hyperproliferation of cells that express CD3 and the presence of characteristic acidophilic masses, like the one labelled TC in photomicrograph B. Which of the following best characterizes these masses?



- A. Located within the cortex of the organ
- B. Functionally inactive component of the organ
- C. Consists of epithelioreticular type VI cells
- D. Consists of epithelioreticular type I cells
- E. Phagocytose apoptotic lymphocytes

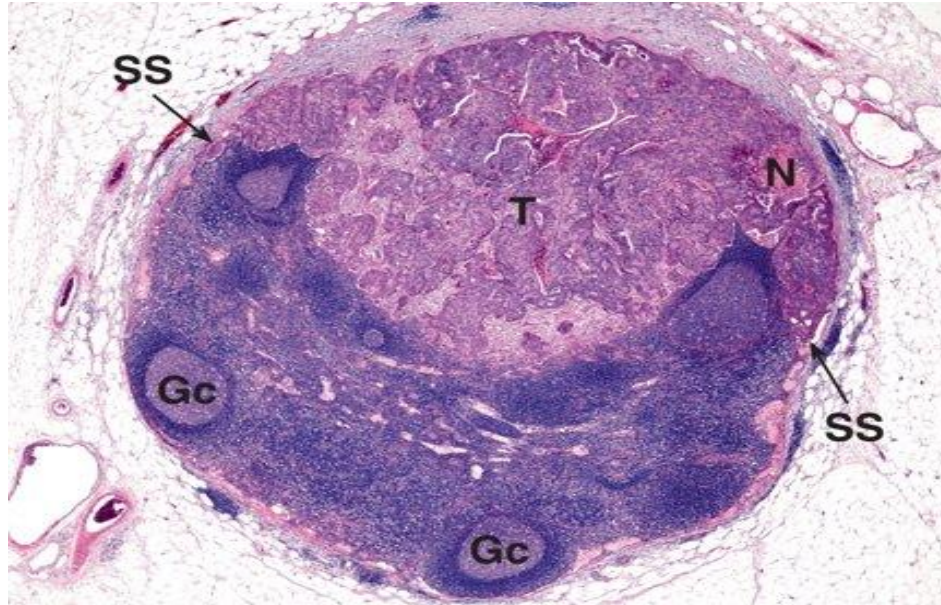
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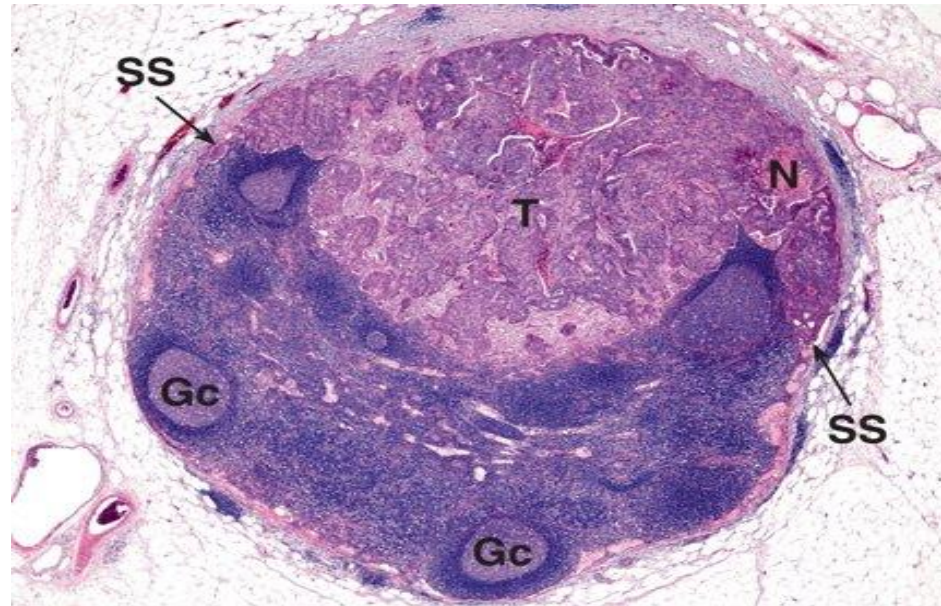
11. A 44-year-old woman comes to the physician because of a 2-week history of swellings in her left armpit. She was diagnosed with left breast cancer 2 months ago and is currently receiving chemotherapy since she refused surgery. Physical examination shows multiple nontender masses in her left axilla. Microscopic examination of a biopsy specimen obtained is shown in the attached photomicrograph. The physician suspects metastasis to lymph nodes. Which of the following best identifies the location where the metastatic cells would most likely be found?

- A. Medulla
- B. Deep cortex
- C. Superficial cortex
- D. Lymphoid follicles
- E. Subcapsular sinus
- F. Trabecular sinus
- G. Efferent lymph vessels



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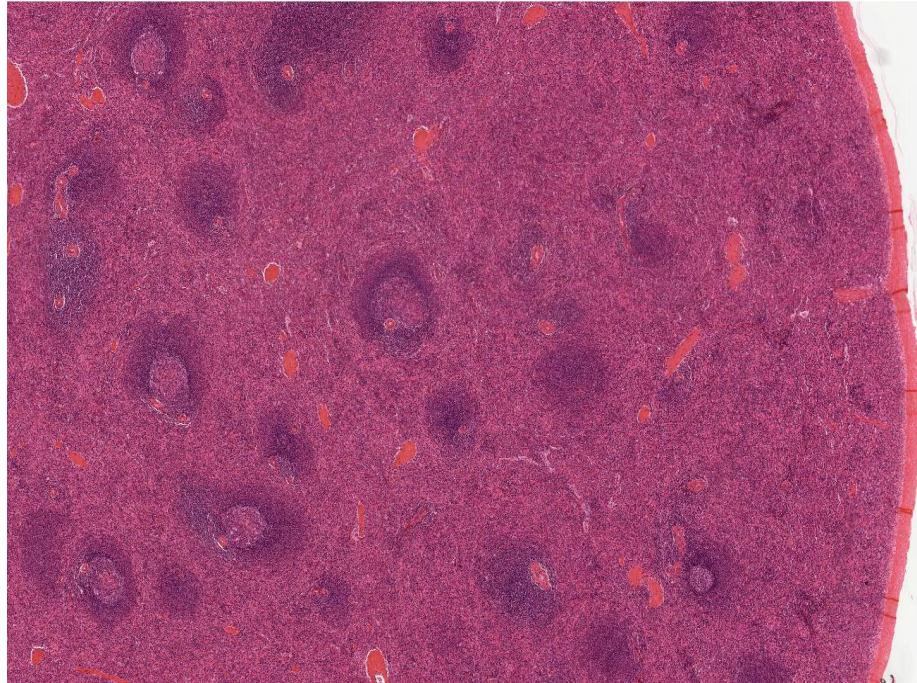
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SOM.MK.1.BPM1.3.CPR.1.HCB.1234 Explain the significance of the subcapsular sinus in cancer metastasis through a lymph node.

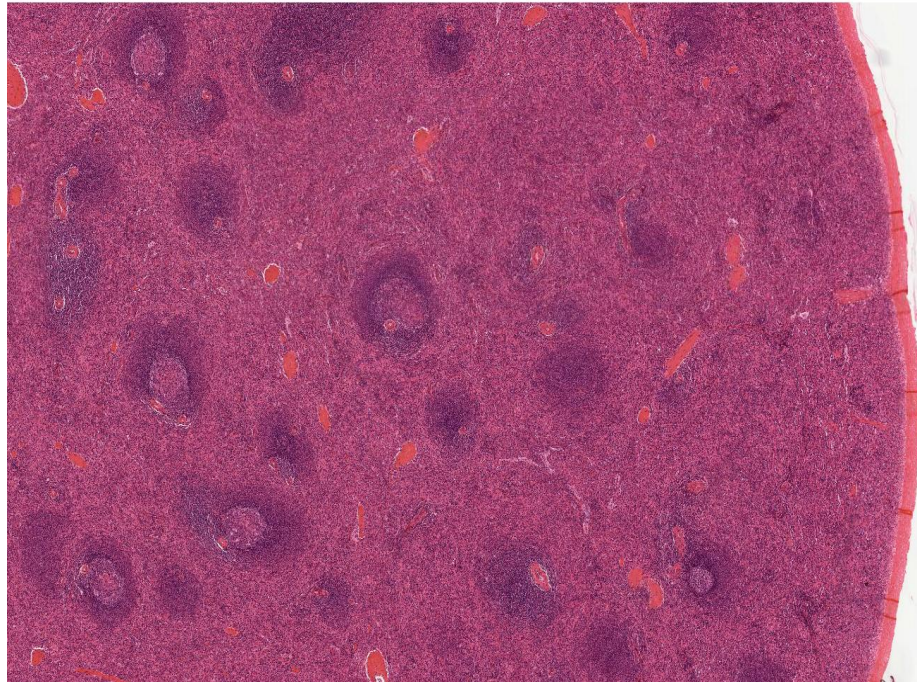
12. A 6-year-old girl is brought to the physician by her mother because of a 3-day history of abdominal pain. She was diagnosed with Sickle Cell disease 2 years ago. Her blood pressure is 90/55 mm Hg, and her pulse is 110/min. Physical shows pallor and yellow discoloration (jaundice) of her mucous membranes, and an enlarged spleen. The physician suspects sequestration (trapping) of blood within the abdominal organ shown in the attached photomicrograph. Which of the following is the most likely location where the blood is trapped?

- A. Periarteriolar lymphatic sheath
- B. Splenic nodules
- C. Marginal zone
- D. White pulp
- E. Red pulp



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13. A 2-year-old girl is brought to the physician by her mother because of a 2-day history of sneezing, runny nose, decreased appetite, and a fever. Her temperature is 38°C (100.4°F). Her other vital signs are within normal limits. Physical examination shows mild wheezing at both lung bases. The physician suspects airway inflammation involving the terminal bronchioles. Which of the following histological features best characterizes these structures?

- A. Walls interrupted by outpocketings
- B. Discontinuous layer of smooth muscle
- C. Presence of type II pneumocytes
- D. Presence of Club cells
- E. Presence of respiratory epithelium

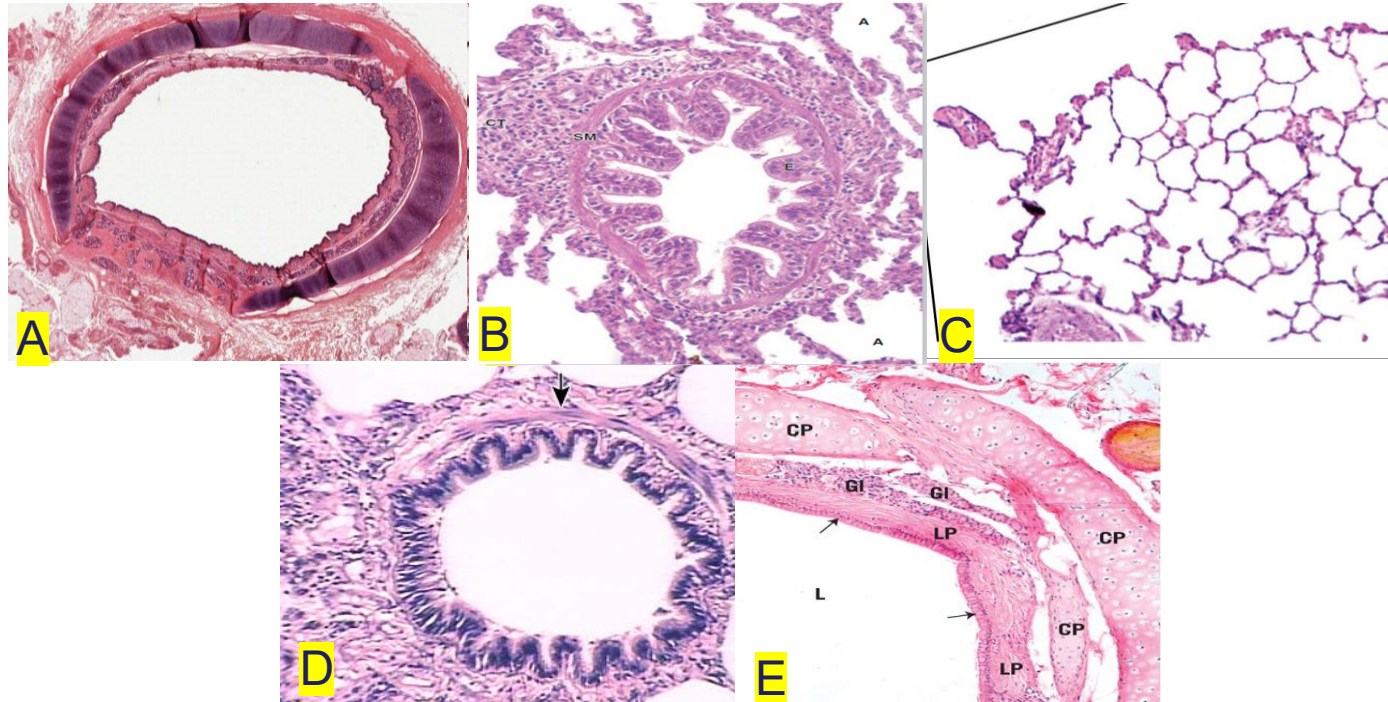
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SOM.MK.1.BPM1.3.CPR.1.HCB.1266 Describe the microscopic appearance of the structure of bronchioles.

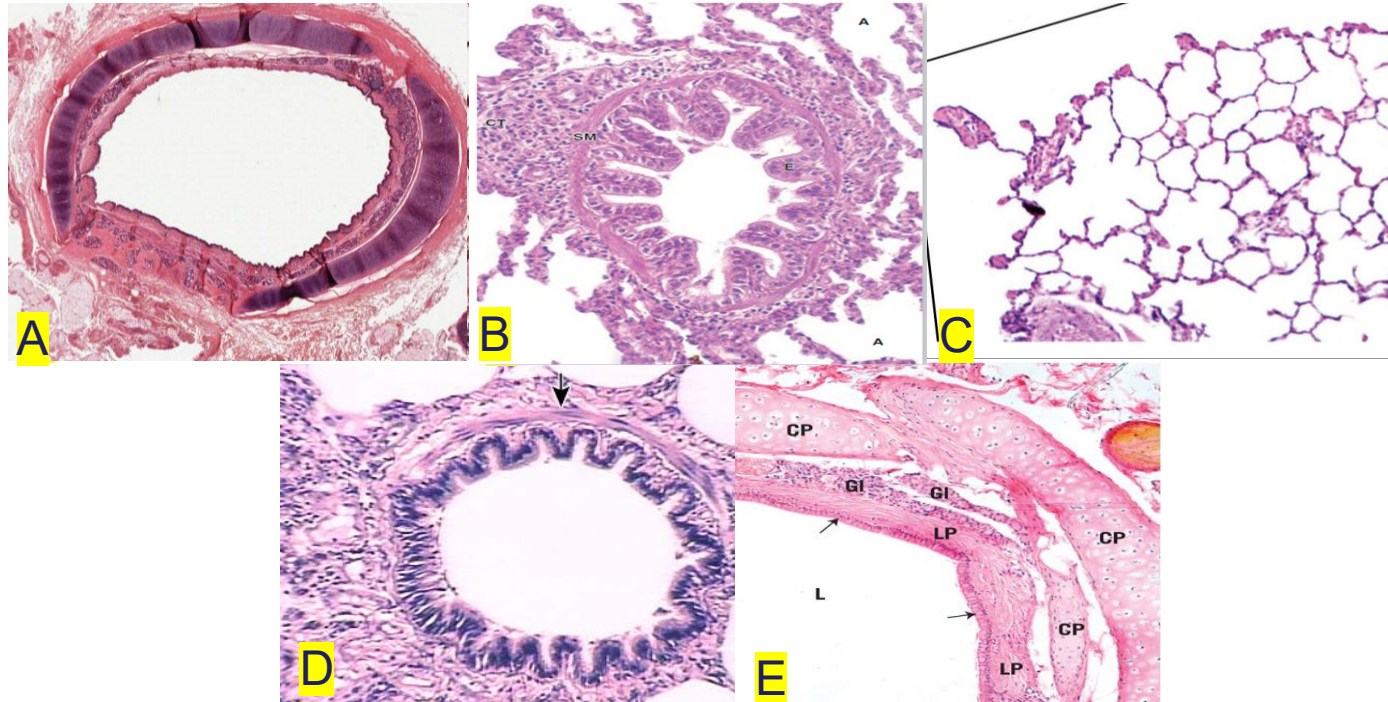
14. A 58-year-old woman comes to the physician because of a 2-week history of shortness of breath, fatigue, and coughing up blood. Her blood pressure is 102/72 mmHg. Physical examination shows pale mucous membranes and fresh blood in a sputum sample. The physician suspects an acute bronchitis with the involvement of the intrapulmonary bronchi. Which of the following photomicrographs best identifies these structures?

- A. A
- B. B
- C. C
- D. D
- E. E



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- A. A
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- D. D
- E. **E**



15. A 49-year-old man comes to the physician because of a 3-month history of hoarseness. He has a 20 pack-year history of cigarette smoking. His vital signs are all within normal limits. Physical examination shows no abnormalities. Laryngoscopy shows a 1-cm mass on one of the laryngeal tissue folds superior to the vocal folds. Which of the following best characterizes the affected tissue fold?

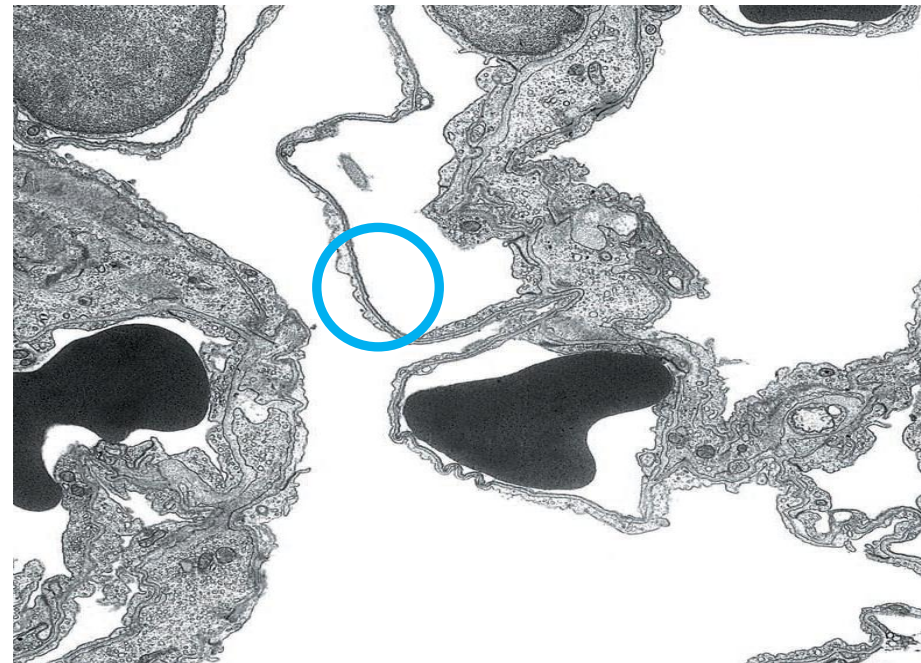
- A. Involved in detoxification
- B. Involved in surfactant production
- C. Simple cuboidal epithelium with club cells
- D. Ciliated, pseudostratified columnar epithelium
- E. Stratified squamous nonkeratinized epithelium

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16. A 24-year-old man comes to the emergency department because of a 2-day history of difficulty breathing. He has a history of uncontrolled type-2 diabetes mellitus. Physical examination shows decreased air entry at both lung bases. A chest x-ray shows small pleural effusion and a mass at the right lung base. Laboratory studies shows increased neutrophil counts in blood and sputum samples. He dies despite appropriate medical care. Microscopic examination of tissue samples collected from autopsy show increased thickness of the tissue indicated by the circle in the attached electron micrograph. Which of the following best identifies a normal function of this structure?

- A. Regeneration of type I pneumocytes
- B. Regulation of the diameter of bronchioles
- C. Detoxification of inhaled air
- D. Contributes to the mucociliary escalator
- E. Forms the site of the blood-air barrier



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